

Beyond Answer Confidence: A Controlled Audit of Self-Knowledge in a Black-Box Decision Model

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Abstract

Decision models return probabilities intended for routing, abstention and automated action. Calibration makes those probabilities useful on average, but does not establish whether low confidence reflects chance or missing knowledge, nor whether confidence falls when a model moves beyond what it knows. We audit this distinction in Jev, a decision model, with over 15 public datasets and 6 generated task families, with paired interventions that vary the information supplied for a fixed item. Jev’s confidence is calibrated on familiar closed-choice tasks but fails as an indicator of missing knowledge: with no answer-relevant information it assigns up to 0.80 to a salient option, and on news beyond an observed knowledge boundary it exceeds accuracy by 0.21–0.33, a gap that recalibration on earlier months does not close. Targeted yes/no questions give sharper readouts of the case: whether an outcome is settled (AUROC 1.00) and whether the evidence suffices (0.95, against 0.85 for confidence on the same items). Asking whether Jev knows the answer appears to flag fabricated entities and post-boundary news (0.91), but with realistic names or with dates removed it shows no advantage over answer uncertainty. Black-box knowledge audits therefore need explicit controls for surface cues. Code: <https://github.com/Syntheme/beyond-answer-confidence>.

1 Introduction

Recently, there has been a surge in interest in decision-focused AI models. This class of models returns typed probability distributions rather than free-form text. These probabilities are intended to support consequential decisions: accept an answer, abstain, route the request to another system, or trigger human review. Their usefulness therefore depends not only on average calibration, but on whether they remain informative when the model lacks the knowledge needed for a particular case.

Calibration alone does not provide that guarantee. A calibrated probability of 0.5 may describe a fair coin or an unknown fact, even though the first reflects chance and the second missing knowledge. More generally, the split of a first-order predictive distribution into aleatoric and epistemic uncertainty is not determined by that distribution [2, 6, 45]. Consequently, epistemic uncertainty cannot be identified by inspecting one output distribution in isolation.

It can, however, be audited behaviourally. We can hold an item and its ambiguity fixed while varying the information available to the model; compare fixed unknown facts

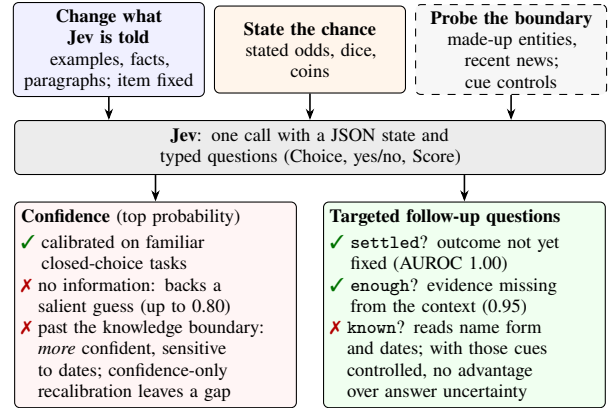


Figure 1: Overview. We change what Jev is told for a fixed item, state the chance of an outcome, or probe the knowledge boundary (dashed: observational; cue controls use realistic names and remove dates), and read two outputs from the same call. Confidence predicts correctness on familiar closed-choice tasks but does not show what Jev does not know; questions about the case detect gaps in it, while “do you know?” mostly reads surface cues (table 1, fig. 3).

with future random outcomes having the same ideal answer distribution; and ask separate questions about whether an outcome is settled, whether the supplied evidence suffices, or whether the model is familiar with the subject. These interventions test whether uncertainty responds to missing information without assuming that the model internally represents an epistemic–aleatoric decomposition.

We apply this audit to Jev, a decision model launched recently. Jev returns probabilities over allowed answers and a derived confidence score. Its authors state that higher confidence predicts higher accuracy and attributes this behaviour to reinforcement learning for calibrated decisions [3, 41]. We distinguish that calibration claim from a stronger operational question: *does Jev’s top-answer probability remain informative when the knowledge required to answer is absent?*

Main finding. Jev’s confidence predicts correctness on familiar material but does not identify why a case is uncertain. It remains high beyond an observed knowledge boundary and selects salient answers when the input provides no basis for choosing one. Targeted questions asked on the same call give direct readouts that confidence provides only indirectly or with the wrong orientation: whether the outcome is settled, whether the evidence is sufficient, and whether the subject is familiar (fig. 1, table 1). Controls for surface

cues separate these: the first two, which describe the case, hold on the tested contrasts; the third, which concerns the model’s own knowledge, is largely explained by unfamiliar-looking names and recent dates: it provides no evidence of memory-gap detection beyond confidence once the measured cues are controlled. A model’s ability to assess the case it is shown should therefore not be mistaken for an ability to identify gaps in its own knowledge.

Contributions.

- **Controlled information interventions reveal where confidence fails.** Jev’s top-answer probability is calibrated on familiar closed-choice tasks, but favors salient options without answer-relevant information and becomes overconfident beyond an observed knowledge boundary; recalibrating on earlier months leaves a large post-boundary gap.
- **Surface-cue controls expose the limits of apparent self-knowledge.** Asking whether Jev knows the answer looks like a strong readout on the original contrasts, but text-only baselines, realistic names and date removal show that name form and dates carry most of that signal.
- **Questions about the case remain useful.** Asking whether an outcome is settled and whether the supplied evidence suffices gives direct, correctly oriented readouts of the tested contrasts (confidence separates them only through mode-seeking or less sharply); an explicit “not known” option helps when many questions lie beyond the boundary.
- **A behavioural audit for black-box decision models.** We combine paired interventions that vary supplied information while holding the item fixed, generated tasks with known ideal distributions, and separate queries about the case, evidence and model familiarity. The audit comprises about 575,000 calls across 15 public datasets and 6 generated task families, with text-only baselines and cue-controlled variants for the follow-up questions.

2 Related work

Epistemic and aleatoric uncertainty. Epistemic uncertainty is the learner’s lack of knowledge of $p^*(y \mid x)$, aleatoric uncertainty its intrinsic randomness [16]. The entropy/mutual-information split is contested [7, 34, 45], second-order losses do not incentivise faithful epistemic estimates [6, 20], provable decomposition needs higher-order supervision [2, 17], and even white-box methods disentangle the two poorly [30]. A model that returns one distribution per query offers nothing to decompose, so we test behaviour.

Black-box tests. We build on sensitivity to irrelevant context [36]; knowledge boundaries via popularity [28], fabricated entities and unanswerable questions [4, 22, 49]; temporal cutoffs [10]; self-evaluation [21, 37, 47]; iterative prompting [1]; and human label distributions [5, 32].

Closest work. LM probabilities deviate from stated odds in coin and dice contexts [27]; token and verbalised probabilities can disagree [44]; and models that describe a fair coin do not behave like one [46]. Concurrent work tests fabricated entities and post-cutoff events across many LLMs [35]. For Jev, early third-party tests compared its multiple-choice and yes/no outputs on hidden fair draws [15], reported its fair-die preference [38], and measured calibration on intent benchmarks [19]. Concurrent papers evaluate Jev as an evaluation judge, where its confidence tracks correctness well enough to route uncertain cases to a stronger model [25], and for semantic choices in scientific pipelines [11]. Those works ask whether confidence predicts correctness; we ask whether it reports missing knowledge. We differ in holding the aleatoric part fixed by design, in tasks with a known ideal for both kinds of uncertainty, and in separating what different follow-up questions measure.

3 Setup

The model. Jev takes a JSON *state*, natural-language instructions and an allowed answer type [39]: a **Choice** returns probabilities over up to 255 named options, a **Noul** (Jev’s yes/no type) returns $P(\text{yes})$, and a **Score** returns probabilities over at most ten ordered levels (appendix A shows a request and response). The API exposes only aliases; we requested `jev-latest`, and every response reported `jev-1.13.0` (collection 26–30 September 2026; the surface-cue controls and verification variants were run on 30 September). At the time, input cost \$0.042 per million tokens and output was free; our cost figures are estimates from token counts, not invoices. The vendor’s documentation acknowledges that answers to related questions need not be coherent [42]. Jev is not deterministic (replicate SD ≈ 0.04 near $p = 0.5$), probabilities are quantised to 0.01, and several questions in one call do not change answers beyond replicate noise. We queried every cell three times and analyse the mean distribution $\bar{q}(y) = \frac{1}{3} \sum_{r=1}^3 q_r(y)$, whose top option is Jev’s answer and whose top probability is its confidence; across 500 random selections of one call per item, the 95% SmoothECE ranges stay within 0.010 of the three-call estimates in all 36 calibration sets (appendix B.1).

Metrics. By Jev’s *confidence* we mean the probability it assigns to its chosen answer, the top probability $p_{\max}$. The API’s *confidence* field is a rescaling of the same quantity (close to $(Kp_{\max} - 1)/(K - 1)$ for K options, 0 at uniform and 1 at certainty), so it carries the same information; we use $p_{\max}$ because calibration needs the probability scale. We report top-label SmoothECE [8], the mean gap between confidence and accuracy, AUROC for error or case detection, normalised entropy $\bar{H} = H/\log K$, and total variation TV to an ideal distribution. Intervals are 95% item-clustered bootstrap intervals (month-clustered for dated news). We call confidence “calibrated” when its estimated top-label calibration error is low (SmoothECE ≤ 0.05) on the closed-choice form of a task; this does not imply classwise or subgroup calibration.

Table 1: Each contrast, what it was intended to measure, and what the controls show it measures. AUROC for the follow-up question and, on the same items, for answer uncertainty $1 - p_{\max}$, or for confidence $p_{\max}$ where that is the orientation that separates the contrast (as marked); “wordings” is the range over paraphrased, reversed and adversarial versions. The contrasts are different tasks, so their AUROCs are not directly comparable. Both news rows use the same 3,360 questions (1,680 per period); wordings for *known* were measured on all real subjects (AUROC 0.91 there). Fabricated-versus-real status and period are proxies for what the model knows, not direct labels of it. The cue-controlled rows (look-alike names, dates removed) are exploratory analyses added after the main studies.

Ground-truth contrast	Intended to measure	Validated interpretation	Answer uncertainty	Question	AUROC	Wordings
Fixed but unknown fact vs. future chance event	outcome settled	outcome settled, also inferred from tense alone (1.00)	0.96, only because Jev concentrates on the mode for chance events; ideal distributions identical	settled	1.00	0.99–1.00
Complete vs. incomplete evidence, same paragraph count	evidence sufficient	evidence sufficient, beyond length features (0.60 → 0.94)	0.85 (confidence, higher = complete)	enough	0.95	0.95–0.97
Pseudo-word made-up vs. obscure real person (same relations)	model’s knowledge	name form: text alone 1.00	0.78	known	0.88	0.87–0.97
Look-alike made-up vs. obscure real person	model’s knowledge	equivalent to answer uncertainty ($\Delta = -0.02$)	0.76	known	0.74	—
News past vs. before the boundary	model’s knowledge	explicit dates: date alone 0.997	0.385 (0.615 reversed)	known	0.91	—
same, answer option				“not known”	0.89	—
News past vs. before, dates removed	model’s knowledge	no demonstrated advantage over answer uncertainty ($\Delta = +0.02$)	0.56	known	0.58	—

Five checks. Let $q(\cdot \mid x, \mathcal{K})$ be the answer distribution for input x with side information $\mathcal{K}$. Uncertainty that reports missing knowledge should show: (C1) *reducibility*—evidence that narrows the correct answers lowers entropy; (C2) *honest ignorance*—if $\mathcal{K}$ says nothing about the answer, q is near uniform over the options not ruled out; (C3) *invariance to non-evidence*—text independent of the answer, declared uninformativ where needed, leaves q unchanged; (C4) *calibration at every knowledge level*; and (C5) a *separate readout*—some other query distinguishes missing knowledge from chance. The checks assume our controlled, symmetric task constructions: evidence that provably narrows the correct answers, options exchangeable under no information, and tasks with a defined answer. They form an audit checklist, not a definition: passing them does not prove that a model knows what it does not know; failures identify limitations in the tested operational behaviour, and passing the checklist does not establish access to a general epistemic state.

Studies. Figure 1 summarises the design. The main studies are:

- *Intent dose–response.* Banking77 [9] (77 intents, 770 test items) and CLINC150 [24] (150 intents, 1,500 items), with intents shown as opaque codes (INTENT_00, ...) that carry nothing, 1–8 example utterances, the intent name, name plus 8 examples, matched-length irrelevant text, or examples swapped across codes. Utterance and label stay fixed.
- *Knowledge×chance scenarios.* 480 generated 4-option

questions about fabricated entities in which the answer is stated, unknown with 0–2 options ruled out, or left to a future random draw with stated chances; the ideal distribution is known for every cell.

- *Knowledge boundary.* PopQA by subject popularity [28], 1,600 fabricated-entity questions in PopQA’s templates, and Daily Oracle yes/no news questions from January 2020 to July 2026 [10].
- *Evidence in context.* HotpotQA comparison questions [48] (two options) with zero, one or two supporting paragraphs, Quizbowl [33] clue by clue, and 600 generated policy cases with a fact edited, deleted, contradicted or joined by an irrelevant one.
- *Random devices and answer formats:* dice, coins and cards over all option orders where feasible (sampled orders for the sum of two dice), and stated odds asked as one Choice or as one yes/no question per outcome.

Standard benchmarks, out-of-scope detection, an open decision model (GLiNER2.5-Decide) and a deep ensemble complete the study (appendices C and E). Free-text benchmarks were converted to four-option Choice. Paired input changes support causal statements about how Jev’s outputs respond (though a change may also alter length or salience); comparisons across popularity, dates and benchmarks are observational and serve as convergent evidence.

Follow-up questions. Each call also asks yes/no questions: *settled* (“is the answer already fixed, so that someone with more information could know it?”), *enough* (“do

Table 2: Intent dose–response: accuracy, mean confidence and SmoothECE by what each intent code carries. With swapped examples, accuracy and SmoothECE are scored against the original labels; the near-zero accuracy shows that Jev follows the swapped mapping, not a calibration failure.

Code carries	Banking77 ($K=77$)			CLINC150 ($K=150$)		
	acc	$p_{\max}$	smECE	acc	$p_{\max}$	smECE
nothing	.010	.320	.310	.008	.363	.355
1 example	.681	.828	.170	.911	.904	.024
8 examples	.899	.927	.037	.979	.974	.016
name	.821	.893	.084	.926	.933	.025
name + 8	.909	.939	.039	.981	.978	.015
irrelevant text	.019	.312	.293	.007	.226	.219
swapped examples	.001	.932	.904	.001	.974	.957

Table 3: Top probability when nothing in the input identifies the answer. Where a hidden answer exists, the ideal is uniform over the options not ruled out. [†]No valid answer exists (false premise); the options are real people, so 0.25 is a uniform forced-choice reference, not a calibrated target. [§]The outcome is not yet decided; the contenders are interchangeable made-up names by construction, so 0.25 is the symmetric value.

Setting	Ideal	$p_{\max}$
Opaque codes (Banking77/CLINC150)	.013/.007	.32/.36
same, open model	.013/.007	.017/.010
Fixed but unknown generated fact	.25	.49
Fabricated entities (PopQA templates)	.25 [†]	.52
Winner of a made-up future event	.25 [§]	.76
Fair die, all 720 option orders	.17	.80

the paragraphs contain enough information to answer for certain?”) and known (“do you know the answer for certain, rather than having to guess?”; for dated news, “do you know for certain how this turned out?”).

4 Results

4.1 Confidence: trustworthy when familiar, blind at the edge

Familiar material. Evidence reduces uncertainty (table 2): entropy falls sharply from no information to one example per intent, and with swapped examples Jev follows the swapped mapping (90% / 98% of items), so it uses the evidence it is given. Confidence is calibrated on familiar closed-choice tasks—CLINC150 at every informative level, with examples or names (SmoothECE 0.015–0.025), PopQA in every popularity quintile (0.020–0.033), MMLU-Redux (0.016), TriviaQA (0.031)—and useful for abstention: answering only at $p_{\max} \geq 0.9$ keeps 8,132 of 9,960 TriviaQA questions with 20 errors.

No information: a salient guess (table 3). With opaque codes only, accuracy is at chance but Jev puts a third of its mass on the first numbered code; for a fair die it puts 0.80 on “one” in every option order. The open GLiNER2.5-Decide model [12, 50] is almost exactly uniform on the same inputs, so this is not intrinsic to decision models. Uninformative cues also move the answer: a guess “made with no informa-

tion” raises the guessed option by 0.65.

The knowledge boundary: a confident default (fig. 2; observational evidence). For fabricated entities, where no option is correct, Jev still puts 0.52 on one option (a uniform forced-choice reference would be 0.25). On yes/no news past an observed accuracy change point around late 2024, Jev answers “no” to 92% of questions, against 61% before, although the true share of “no” is 51–52% in both periods (51.2% of 4,640 before, 51.5% of 1,680 after), and its confidence *rises* from 0.75 to 0.82. The dates in the questions (98.7% contain a year) drive much of this: with every temporal expression removed, post-boundary “no” answers fall from 92% to 67%. Some removals change the proposition (“Will the Fed cut rates by the end of March 2024?” loses its deadline) or remove which event is meant, so this is a change in responses, not a validated change in accuracy. Among questions whose original answer is “yes” (1,648), removing dates cut post-boundary “no” answers from 92% to 57%, against 41% to 37% before the boundary; but in a 240-pair sample provisionally annotated by an AI assistant under a written protocol, 30 of the 112 “yes” questions lost event identity, so these are not all answer-safe. The pairs judged answer-preserving show the same direction (92% to 65% after, $n = 37$; 39% to 35% before, $n = 46$; appendix C.2). Removing dates also removes the specificity that identifies an event, so these edits do not separate superficial date-reading from temporal knowledge retrieval. The question’s own date matters, not the date of asking: stating today’s date as just after the event leaves the post-boundary “no” share at 91%. The preference is not tied to the word “no”: Jev verified both a proposed “yes” and a proposed “no” for every item, with both a positive (“is the answer correct?”) and a negative (“is it wrong?”) wording. The two wordings agree on 95–96% of items, and past the boundary both imply “yes” for only 8–10% of questions (true share 48%); a preference for the word “no” would make the negative wording accept both proposals, which happens for 0.1%. This is behavioural evidence of a default towards “it did not happen”, not proof of a belief. Shifts between broad news categories alone do not explain the drop, since accuracy falls within every category. The change point is an accuracy break, not a verified training cutoff. On these yes/no questions Jev’s log loss (0.71) is worse than always answering 50/50 (0.69). Material built to be novel shows the same pattern (MMLU-CF: accuracy 0.78 at confidence 0.91). **Recalibrating confidence does not close this gap.** Recalibrating forward in time, on earlier months only, leaves post-boundary questions 0.19–0.23 overconfident. At equal confidence above 0.6, post-boundary questions are 0.17–0.31 less often right. Because $p_{\max}$ takes only 151 distinct values on this panel, all seen on both sides, recalibration maps can be bounded directly. An exact optimisation over contiguous partitions of the values, which relaxes monotonicity, shows that no order-preserving map can bring the period-conditional calibration error below 0.047 (a bound that need not be attained). Searches over arbitrary groupings improve this only modestly in sample (0.043–0.046) and are worse on held-out months, where the cross-fitted contiguous-partition map leaves post-boundary questions 0.18 overconfident; a map that also knows the period closes both gaps (exploratory, on this news panel; appendix B.4). The differences persist

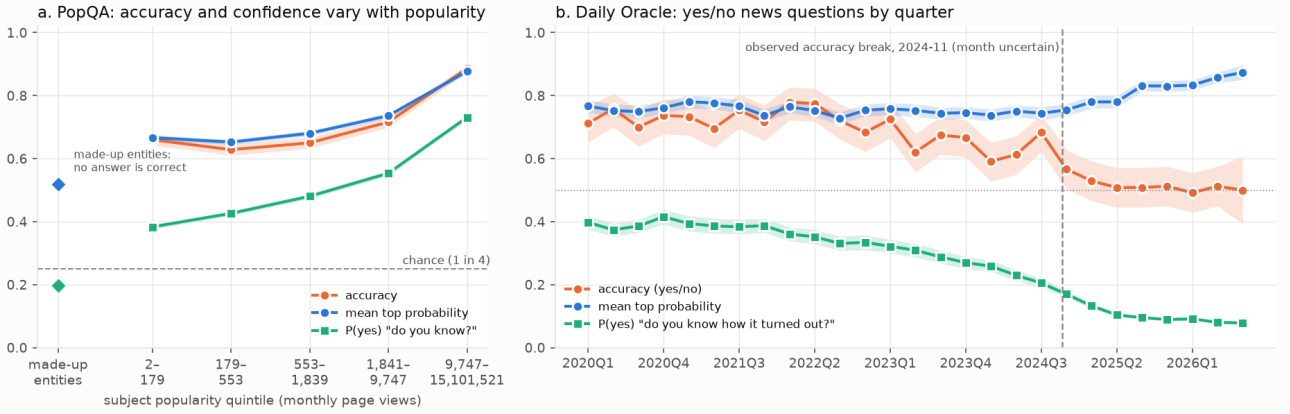


Figure 2: Knowledge boundary. (a) PopQA: confidence tracks accuracy across popularity quintiles, but fabricated entities still get 0.52 on a top option, while $P(\text{known})$ drops to 0.20. (b) Daily Oracle yes/no questions by quarter: after the observed accuracy break (change point, around late 2024) accuracy falls to chance while confidence *rises*; $P(\text{known})$ declines steadily, a signal carried mostly by the dates in the questions (section 4.2).

when the change point is re-selected inside a block bootstrap and at every fixed cutoff from June 2024 to March 2025. Distinguishing the two periods requires information beyond the confidence score in these analyses.

4.2 What targeted questions reveal, and what they do not

Table 1 is the paper’s central result. For gaps in the case, a single yes/no question asked on the same call gives a direct readout, sharper or better oriented than confidence; for gaps in the model’s memory, the apparent signal does not survive controls.

“Is the answer already settled?” When a fact is fixed but unknown and when an outcome awaits a fair random draw, a calibrated answer distribution has the same ideal in both cases, so which kind of uncertainty is at play is not identifiable from that distribution alone. Jev’s actual answer uncertainty does separate them (AUROC 0.96, higher for the fixed facts), but only because Jev concentrates on the mode of a stated chance event, itself a calibration failure (section 4.4): even in the uniform-chance scenarios, where the ideal distributions are identical, its mean confidence is 0.76 on future draws against 0.49 on unknown facts (AUROC 0.89). Read as a statement of knowledge, it points the wrong way. *settled* exceeds it by 0.04 [0.03, 0.06] overall and 0.11 [0.07, 0.16] in the uniform scenarios. The *settled* question separates them directly (mean $P(\text{yes})$ 0.87 vs. 0.13; fig. 5c), including traps that pit meaning against wording (a past random draw is settled; a council decision next week is not) and made-up events whose status must be inferred from tense (“was played last spring” vs. “will be played next spring”) or from dates, provided today’s date is given.

“Is there enough information?” This question gives a direct readout of evidence completeness. On the same two-paragraph HotpotQA conditions it separates complete from incomplete evidence with AUROC 0.95. Jev’s confidence also rises with complete evidence (AUROC 0.85, higher meaning complete), and text length alone reaches 0.65; on the same items enough beats confidence by +0.10 [0.09, 0.11], so the question adds a clearer signal rather than one

confidence lacks. Separately, misleading context reduces accuracy without reducing confidence: with related but non-supporting paragraphs, HotpotQA accuracy falls from 0.84 to 0.75 while confidence stays at 0.83; $P(\text{enough})$ is 0.13 for these and 0.86 with both supporting paragraphs. All conditions with paragraphs show two of them, so paragraph count does not distinguish them (text length is discussed below). When a deciding fact is deleted from a policy case, $P(\text{enough})$ falls from 0.83 to 0.10 and a “cannot tell” option receives 0.91.

“Do you know the answer?” This is the only one of the three about the model’s own knowledge. On the original contrasts it flags fabricated entities (0.91, where $1 - p_{\max}$ reaches 0.75; 0.88 against the same obscure real people used below, where $1 - p_{\max}$ reaches 0.78) and news past the boundary (0.91, where confidence ranks post-boundary questions *higher*), and it beats the strongest self-evaluation baseline we ran, a second-look “is the proposed answer correct?”, on the same items (0.92 vs. 0.80 for made-up entities, 0.91 vs. 0.62 for news); across time it falls steadily from about 0.40 for 2020 news to about 0.10 for 2025–26 news (fig. 2b).

Controlling for surface cues (fig. 3). These contrasts contain cues that require no knowledge. Our fabricated names are pseudo-words: a character n-gram classifier on the question text alone separates them from real subjects perfectly (AUROC 1.00, also against the least popular real ones). Likewise, 98.7% of the news questions contain a year, and the date alone separates the periods (0.997). Controlling these measured cues removes most of *known*’s signal. Against the same 555 obscure real people and relations, look-alike names built from real first names and surnames leave a text-only classifier at chance (0.49), and *known* drops from 0.88 (pseudo-word names) to 0.74 [0.71, 0.77], similar to answer uncertainty (0.76 [0.73, 0.79]). With explicit temporal expressions removed from the news questions, *known* falls to 0.58 [0.55, 0.61] (answer uncertainty 0.56 [0.54, 0.58]); this comparison uses only period membership, so it does not depend on the rewritten questions’ answer labels. A text classifier still separates the masked periods at 0.83, so period information remains in the text; Jev’s familiarity

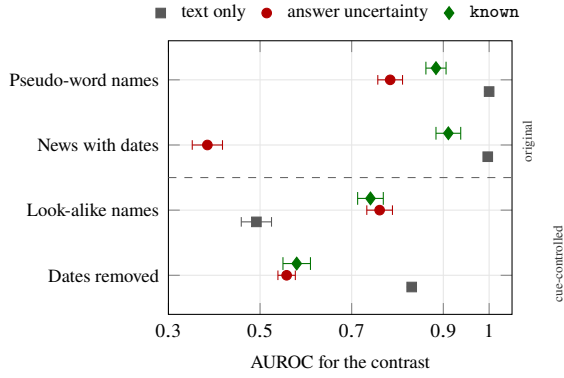


Figure 3: Controlling for surface cues. On the original contrasts (upper two rows) a text-only classifier that never queries Jev separates made-up from real subjects and later from earlier news almost perfectly, and *known* scores 0.88–0.91. With realistic names and with explicit temporal expressions removed (lower two rows), the text classifier weakens and *known* falls to the level of answer uncertainty ($1 - p_{\max}$). Names rows use the same 555 obscure real people as controls (text-only for pseudo-word names: against the least popular real subjects). Answer uncertainty below 0.5 for news with dates means confidence is *higher* after the boundary. Both news rows use the same 3,360 questions (1,680 per period); the date-only text baseline is from the full panel, and the masked-text baseline is month-grouped (0.83). 95% intervals; for masked news the text classifier still uses topic cues. Exploratory analyses, added after the main studies.

answer uses little of it. On the same items, the paired difference $\text{known} - (1 - p_{\max})$ is $-0.02 [-0.05, 0.01]$ for realistic names, equivalent within ± 0.05 (90% interval), and $+0.02 [-0.01, 0.06]$ with dates removed, where equivalence at that margin is not shown but there is no demonstrated advantage; with dates present it is $+0.53$. The equivalence margin, like these controls, was chosen after the main studies. Grouping the text-classifier folds by subject or month preserves the qualitative conclusions (1.00 and 0.50 for names; 0.83 for masked news, against 0.86 ungrouped). *known* does respond to information in the input: when one sentence states the answer about a made-up subject, $P(\text{known})$ is 0.81 with an answer-bearing sentence, 0.20 without a sentence and 0.05 with an unrelated sentence. Within popularity levels or news months it ranks right against wrong answers only weakly (0.48–0.66). On these tests it therefore tracks whether an answer is available in the input or looks familiar; we find no robust additional signal about whether Jev holds the knowledge, although fabricated-versus-real status and period are only proxies for that. By contrast, *enough* is not explained by length. Among HotpotQA conditions with two paragraphs, complete evidence is the *shortest*, so length separates them with AUROC 0.65 in the “shorter means complete” direction; comparing only items within the same 10-word length stratum (a size-weighted mean of within-stratum AUROCs), *enough* keeps 0.95 while length falls to 0.48, and adding *enough* to a gradient-boosted model of length features (question and paragraph lengths) raises AUROC from 0.60 to 0.94. For pairs from the same question within 10 words of each other, *enough* ranks the complete one higher 96% of the time.

Scope. Each question targets a specific gap. *settled* and *enough* judge the input; *known* tracks surface familiar-

ity and supplied evidence, and is a weak detector of individual errors on ordinary questions (TriviaQA AUROC 0.76 vs. 0.96 for $p_{\max}$). *enough* detects incomplete evidence, not wrong answers: Jev often answers correctly without the paragraphs, so accepting answers by *enough* is 2–6 points less accurate than accepting by confidence at the same coverage (thresholds set on held-out questions). The rankings are robust to fourteen rewordings, but absolute levels vary, so thresholds must be set per wording. These AUROCs measure discrimination of curated contrasts; the answers are ranking scores, not validated probabilities of an epistemic state. Ambiguity is flagged only weakly (“single clear answer?” AUROC 0.66).

4.3 Putting the questions to use

Combining. Adding $P(\text{known})$ to confidence raises error-detection AUROC on dated news from 0.61 to 0.66, and a recalibration map that uses both cuts the post-boundary gap from 0.30 (raw) and 0.24 (confidence alone) to 0.16; a map that knows the period closes it (appendix B.4). On ordinary questions $P(\text{known})$ adds nothing, because there confidence already works.

A “not known” option puts the *known* idea into the answer itself. Jev chooses it for 93% of post-boundary news questions, so it answers only about 7% of them, and for 36% of earlier ones; confident errors nearly vanish among the questions it answers, a count that depends on the format, since the third option changes the probability scale. At matched coverage it beats answering by confidence when half the questions lie past the boundary ($+0.05$ accuracy), is inconclusive at the 27% share of our date panel ($+0.01 [-0.00, 0.04]$), and is worse when such questions are rare ($-0.03 [-0.05, -0.01]$ at 5%, an interval close to zero; at 10% the difference is inconclusive; fig. 4b). It routes boundary questions rather than ranking errors, and it beats thresholding *known* at every mix. These questions contained their dates, which likely help the option as they help *known*. A cheap rule gets close: skipping the questions whose own latest date is most recent, at the option’s coverage and using Jev’s ordinary answers, trails the option by only 0.01–0.02 at every mix (at the panel’s share, $+0.02 [0.00, 0.04]$ in the option’s favour; coverage and error rates in table 9).

Stating that context may be irrelevant restores HotpotQA accuracy with misleading paragraphs ($0.75 \rightarrow 0.84$) and calibration (SmoothECE 0.02). A second-look check (“is the proposed answer correct?”) reduces estimated pooled calibration error on the balanced news set of 1,680 questions from each period (SmoothECE $0.18 \rightarrow 0.03$), which does not by itself establish calibration within each period, but makes easy sets underconfident. Giving today’s date alone barely helps.

4.4 For stated chances, ask one question per outcome

The same lesson holds for aleatoric uncertainty. Told that a future draw yields outcomes with chances 40, 30, 20 and 10%, Jev’s Choice puts 0.98 on the 40% outcome (fig. 5a), and instructing it to report chances changes nothing. Asked one outcome at a time, the yes/no probability is within

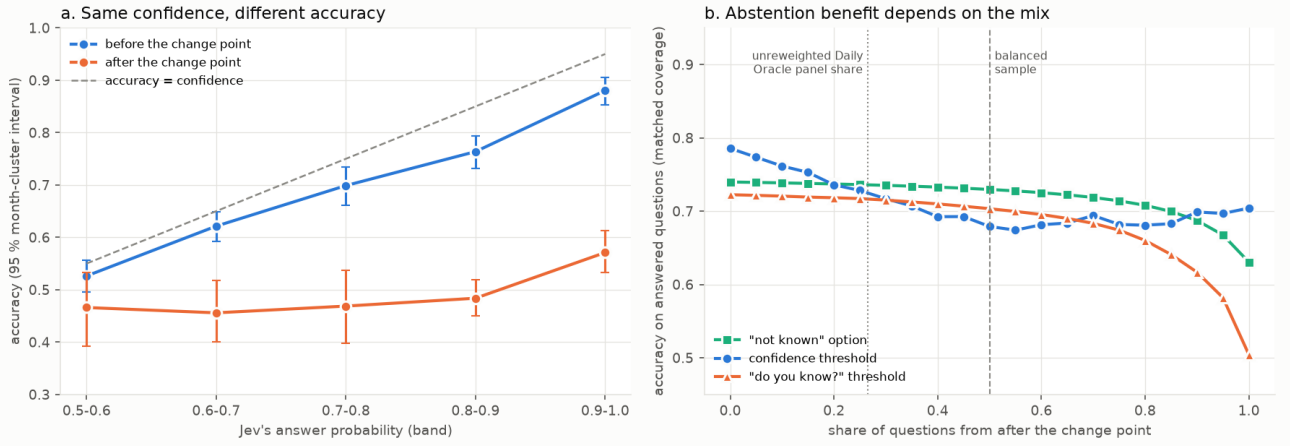


Figure 4: Daily Oracle yes/no questions around the change point (2024-11). (a) Accuracy at equal confidence (95% month-cluster intervals): lower and nearly flat after the change point, so a common map of confidence cannot match accuracy at each value on both sides. (b) Accuracy at matched coverage as the share of post-change questions varies: a “not known” option beats a confidence threshold when such questions are common and beats thresholding “do you know?” at every mix. The dotted line marks the share in our date panel, not a deployment prevalence.

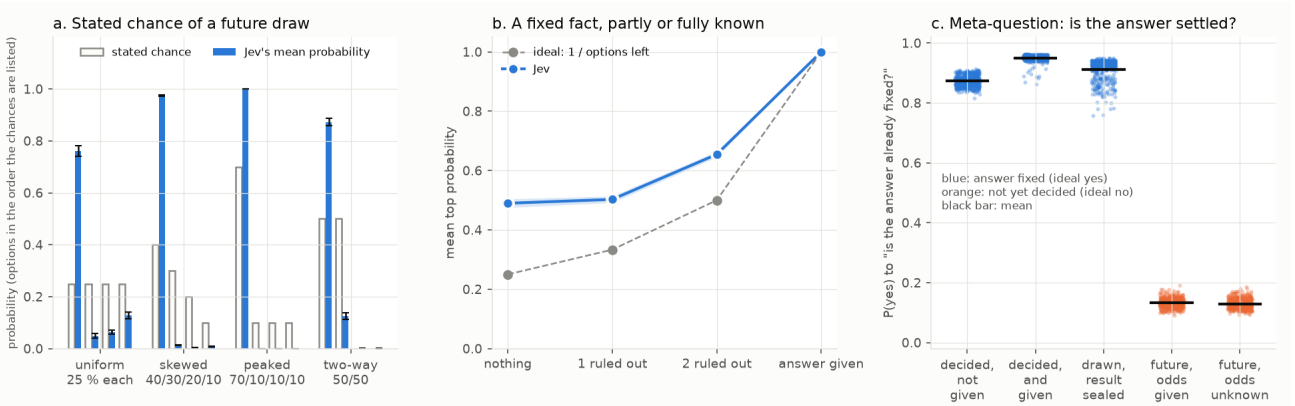


Figure 5: Knowledge×chance scenarios (480 generated questions). (a) Stated chances of a future draw vs. Jev’s Choice probabilities: nearly all mass on the likeliest outcome, ties broken towards the outcome mentioned first. (b) A fixed fact, progressively revealed: right direction, too confident throughout. (c) The settled question separates fixed-but-unknown answers from future draws perfectly.

Table 4: Mean reading by stated chance (480 scenarios).

Stated	0	.1	.2	.3	.4	.7	MAE
Yes/no per outcome	.01	.09	.16	.25	.36	.68	.030
Score per outcome	.05	.13	.22	.34	.45	.74	.028
Instructed Choice	.00	.00	.00	.01	.99	1.0	.230

0.030 of the stated chance on average (table 4); the readings sum to 0.89, and rescaling them to one lowers the error to 0.012. Our Choice prompts ask for the realised outcome (e.g. “What is the colour of the guild’s banner at the coming festival?”, instruction “Answer the question using the facts given”), and the vendor describes a Choice as returning the full probability distribution over the options, with probabilities optimised against outcomes [40, 43]. Read that way, 0.98 on a 40% outcome is badly miscalibrated. Rephrasing does not change it: asking which outcome “will come up”, which is “most likely”, or which one “you win if you pick” gives the same concentration on the mode (loaded die: 1.00 on its 50% face; sum of two dice: 0.98–1.00 on 7). Only asking for “the probability of each” moves part-way towards

the stated odds (loaded die 0.66; two dice 0.54 on 7, stated 0.17). Under no tested wording did a Choice distribution behave as an outcome distribution, and ties fail too: 0.80 on “one” for a fair die, following labels rather than display positions, which survives averaging over option orders [52]. For outcome probabilities, ask one yes/no question per outcome.

5 Discussion

Jev’s confidence and the follow-up questions play different roles. Confidence is the probability Jev assigns to its chosen option: calibrated on the familiar closed-choice benchmarks we tested, where it supports abstention (e.g. at $p_{\max} \geq 0.9$), although we did not evaluate deployment utility under other prevalences, costs or shifts. It is not a readout of what the model does not know—when nothing decides the answer, Jev still names one, chosen by salience, and past its knowledge boundary its confidence goes up, with substantial sensitivity to the dates in the questions. Targeted questions provide more direct or sharper readouts of partic-

ular case properties than answer confidence: whether an outcome is settled and whether the evidence supplied suffices. Asking Jev whether it knows the answer looked like a read-out of its own knowledge, but controls show that it largely tracks unfamiliar-looking names, explicit dates and supplied evidence; once the measured cues are removed it shows no demonstrated advantage over confidence. For Jev, gaps in the case are therefore better *queried* than read from a single distribution, whereas the familiarity question provides no evidence of memory-gap detection beyond confidence once the measured cues are controlled; our labels of such gaps are proxies. This is a case study of one model; whether the same holds for other decision models is untested. We do not claim these questions recover a general epistemic state; each catches the gap it asks about.

Practical guidance. On familiar, in-scope closed-choice questions like those tested, Jev’s confidence can be used for abstention; a single call’s aggregate calibration is close to that of three. Add *settled*, enough where outcomes may be undetermined or context incomplete; do not rely on *known* to detect gaps in the model’s memory, and set thresholds per wording on held-out data. Evaluate a “not known” option at the expected input mix and error costs; in our news comparisons it improved matched-coverage accuracy only when later-period questions were common: it answers few such questions (about 7% past the boundary), is inconclusive at our panel’s mix and costs accuracy when they are very rare (5%), and a simple rule that skips questions with the most recent dates comes within 0.01–0.02 of it; give the current date when asking whether an outcome is settled, and state that retrieved context may be irrelevant. To recover explicitly stated outcome probabilities, ask one yes/no question per outcome and rescale the readings to sum to one. This is validated against the stated odds, including expected proper scores under the stated process (appendix C.3); for real-world outcomes it is untested.

6 Conclusion

On the familiar closed-choice benchmarks we tested, Jev’s confidence tracks how likely an answer is to be right, but it does not show what the model does not know: with nothing to go on it backs a salient guess, and past its knowledge boundary it grows more confident in a way that confidence-only recalibration does not repair. Targeted questions on the same call—is the outcome settled, is there enough information—discriminate gaps in the case on the tested contrasts, while asking whether it knows the answer mostly reads surface cues. Gaps in the case can be asked about; gaps in the model’s knowledge remain hard to see from outside, and auditing them requires controls that remove the cues a model can read instead.

Code availability

The code for the experiments and analyses is available at <https://github.com/Syntheme/beyond-answer-confidence>.

Limitations

Scope. This is a case study of one version of one closed model; no other system was run through the full protocol, so we make no claim about decision models or LLMs in general. **Surface cues.** The follow-up questions are evaluated on curated contrasts. For *known*, text-only baselines, realistic names and date removal show that surface cues carry most of its signal; a text classifier at chance does not certify that no cue remains. *settled* is by design a judgement about the input, and *enough* was checked against text length and length features only. **Date removal.** Removing temporal expressions can change a question’s proposition (e.g. by dropping a deadline) or remove which event is meant, so we report response changes, not validated accuracy; the provisional labels of 240 pairs come from a single annotator (an AI assistant following a written protocol). **Observational boundary.** The knowledge boundary is an accuracy change point in dated news, not a verified training cutoff, and topic or style may change with it. **Forced choice.** Most benchmarks were converted to multiple choice. Fabricated-entity questions have no valid answer (a false premise), so their 0.25 is a uniform forced-choice reference; made-up future events have a defined predictive target that is symmetric by construction; and fixed but unknown generated facts have a hidden answer with a uniform prior. **Probability semantics.** The vendor does not specify whether a Choice models the realised outcome or the best answer. **Scores, not probabilities.** The follow-up answers are validated as rankings, not as calibrated probabilities of an epistemic state. **Decision value.** Benefits of the “not known” option depend on how often boundary questions occur. Some answer-filter comparisons use thresholds chosen on held-out questions, but cost-based utility and a complete deployment routing policy were not evaluated. **Exploratory analyses.** The surface-cue controls, recalibration bounds and robustness analyses were added after the main studies and reuse the same dated-news panel; the temporal findings have not been confirmed on a fresh panel.

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A Example request and response

One call from the knowledge×chance scenarios (option “blue” ruled out, so the ideal Choice answer is 1/3 on each remaining option); the response is shown as returned, with usage counts omitted.

```
state: {"facts": [
  "Voskin is mentioned in 31 old records.",
  "The name Voskin comes from an old word meaning 'bridge'.",
  "The colour of the Voskin guild's banner at the coming
  festival has already been decided.",
  "The colour of the Voskin guild's banner at the coming
  festival is not blue."],
"question": "What is the colour of the Voskin guild's banner
at the coming festival?"}
questions: {
  "answer": {"type": "choice",
    "instructions": "Answer the question using the facts given.",
    "criteria": {"red": null, "green": null,
      "blue": null, "black": null}},
  "settled": {"type": "noul",
    "instructions": "Is the answer to the question already fixed
at this moment, so that someone with more information than
is given here could know it for certain?"}}

response: {"model": "jev-1.13.0", "answers": {
  "answer": {"type": "choice", "choice": "red", "confidence": 0.46,
    "probabilities": {"red": 0.60, "green": 0.23,
      "black": 0.15, "blue": 0.02}},
  "settled": {"type": "noul", "noul": 0.89}}}
```

The request also carried a second follow-up question (“do the facts determine the answer with certainty?”, $P(\text{yes}) = 0.04$), omitted here.

B Statistical methods and robustness

Intervals. Intervals are 95% item-clustered bootstrap intervals with 2,000 resamples; dated news uses months as clusters, and time-dependent analyses use period-stratified 6-month moving blocks. For paired AUROC comparisons, resampling is by original question (HotpotQA) or generated scenario, retaining all associated conditions and all compared scores. We report effect sizes and intervals rather than pass/fail tests.

B.1 A single call

Over 500 random draws of one of the three cached calls per item, single-call SmoothECE stays within 0.010 of the three-call value (95% range) and accuracy within 0.01 in all 36 calibration sets. The point estimate falls on the same side of 0.05 as the three-call value in every draw except SimpleQA Verified (three-call 0.051; below 0.05 in 11% of draws).

Table 5: SmoothECE of the mean of three calls and of one random call per item (500 draws).

Set	<i>n</i>	mean of 3	single, 2.5–97.5%	estimate < 0.05
Banking77, 8 examples	770	0.037	0.031–0.040	100%
CLINC150, 1 example	1,500	0.025	0.021–0.026	100%
PopQA	14,267	0.021	0.021–0.025	100%
Daily Oracle, after change point	1,680	0.305	0.304–0.310	0%
TriviaQA	9,960	0.031	0.030–0.032	100%
SimpleQA Verified	1,000	0.051	0.048–0.061	11%
MMLU-Redux	5,330	0.016	0.015–0.018	100%
MMLU-CF	10,000	0.145	0.144–0.147	0%

B.2 Coverage of the SmoothECE intervals

Confidences were drawn from four real Jev confidence pools at their real sizes, with correctness Bernoulli($\min(1, \max(0, p - \delta))$) and δ set so that the population SmoothECE is 0, 0.03, 0.05 or 0.07 (400 datasets per setting). At a true value of exactly

0.05, the one-sided 95% upper bound falls below 0.05 in 1–6% of datasets for the plain percentile bootstrap (nominal 5%), 2–9% for a bias-recentred bootstrap, and 13–28% for the basic bootstrap and half-sample subsampling. We therefore report percentile bounds. The 0.05 threshold is used only descriptively. The estimator is biased upward by 0.010–0.025 for perfectly calibrated data, so small-sample values of 0.02–0.03 largely reflect that floor. Only one shape of overconfidence was simulated.

B.3 The knowledge change point

The change point is the month that best splits monthly yes/no accuracy into two means. Re-selecting it in 2,000 resamples, a drop of at least 0.05 is always found (2024-11 in 49% of resamples, mostly 2024-01 to 2024-10 otherwise); the post-change gap between confidence and accuracy is 0.21–0.33 and the rise in confidence +0.02 to +0.07. Every fixed cutoff from 2024-06 to 2025-03 gives a gap of 0.26–0.32. With flexible time trends (natural splines, HAC intervals), $P(\text{known})$ shows no step at the change point (−0.014 [−0.033, 0.005]); it declines steadily.

B.4 Recalibration at matched confidence values

On the Daily Oracle yes/no questions, $p_{\max}$ takes 151 distinct values, all observed in both periods. A recalibration map assigns an output level to each value; for a level c shared by a group of values, its period-conditional error is $n_{\text{pre}}|c - a_{\text{pre}}| + n_{\text{post}}|c - a_{\text{post}}| \geq \min(n_{\text{pre}}, n_{\text{post}})|a_{\text{pre}} - a_{\text{post}}|$, computed on the group’s pooled accuracies a . Merging values can cancel differences between periods, so we evaluate three classes of map (table 6). The value-wise bound (each value its own level) is 0.066; over all contiguous partitions of the values, solved exactly by dynamic programming without requiring the levels to increase, the minimum is 0.047, which is therefore a lower bound for every monotone map; random sets of 21 months give 0.005 (descriptive $p = 0.005$). A local search over arbitrary groupings finds 0.043–0.046. Cross-fitted on held-out months, every class leaves a large error, and the cross-fitted contiguous-partition map leaves post-change questions 0.18 overconfident. Recalibrating forward in time on earlier months only leaves +0.19 after the change point with an expanding window and +0.23 with a map frozen at the change point (+0.17 to +0.31 by month). A period-aware isotonic map closes both gaps, and a logistic model with period terms gives $\chi^2_2 = 281$. The confidence-band intervals depend on the procedure: with the change point fixed at 2024-11 and months resampled within each period (table 7), the lowest band’s interval includes zero (−0.014 to 0.139); re-selecting the change point in every period-stratified 6-month-block resample, all five exclude zero (lowest band 0.015 to 0.123).

Optimisation. Let distinct confidence values be sorted, and let a map group them and assign each group one output level c . For a group with $n_{\text{pre}}, n_{\text{post}}$ items and pooled accuracies $a_{\text{pre}}, a_{\text{post}}$, the smallest achievable error $n_{\text{pre}}|c - a_{\text{pre}}| + n_{\text{post}}|c - a_{\text{post}}|$ over $c \in [0, 1]$ is $\min(n_{\text{pre}}, n_{\text{post}})|a_{\text{pre}} - a_{\text{post}}|$; the total is summed over groups and divided by all 6,320 questions. The interval result minimises this total over every partition of the sorted values into contiguous groups, by dynamic programming ($E_j = \min_{i < j} [E_i + \text{cost}(i, j)]$), without requiring the levels to increase. Since every monotone map’s level sets are contiguous, this relaxation is a lower bound on the error of every monotone map. The comparison distribution repeats the optimisation after drawing 21 of the 79 months at random as “after” (200 draws). It treats months as exchangeable, ignoring drift, serial dependence and the selection of the change point, so its p -value is descriptive; a temporally structured null (contiguous placebo boundaries or moving blocks with the boundary re-selected) is not reported; cross-fitting splits months within each period in half (50 random splits, used in both directions). A fitted map is applied as follows: the partition is chosen on the training months using their period labels (the objective is period-conditional), each group’s output level is the training accuracy of its larger period (the minimiser of the group cost; ties go to the earlier period), and a held-out value whose group has no training items receives the overall training accuracy; held-out error is then scored on the fitted groups. This is a partition-conditioned discrepancy: two groups that receive the same output level are scored separately, so it need not equal calibration error after equal output levels are pooled. The map itself takes only confidence as input. The cross-fitted error averages over both periods, while the post-change gap is the mean signed overconfidence after the change point, so the two numbers measure different things. These are post-hoc results for this news panel.

Table 6: Period-conditional calibration error of confidence-only recalibration maps on dated news.

Maps allowed	In-sample	Random months (descriptive), mean / 95th pct	Cross-fitted mean (range)
Value-wise (each value its own level)	0.066	0.034 / 0.038	0.170 (0.147–0.191)
Contiguous partitions (exact; lower bound for monotone maps)	0.047	0.005 / 0.013	0.057 (0.046–0.074)
Arbitrary groupings, 2–20 levels (search)	0.043–0.046	—	0.093–0.099

Table 7: Accuracy by answer-probability band before and after the change point (month-cluster intervals).

$p_{\max}$ band	n before	acc. before	n after	acc. after	Difference [95% CI]
0.5–0.6	988	0.526	206	0.466	0.060 [−0.014, 0.139]
0.6–0.7	943	0.621	204	0.456	0.166 [0.100, 0.228]
0.7–0.8	777	0.699	224	0.469	0.230 [0.152, 0.306]
0.8–0.9	821	0.764	370	0.484	0.280 [0.236, 0.327]
0.9–1.0	1,111	0.880	676	0.571	0.309 [0.260, 0.354]

C Additional results

C.1 Surface-cue controls for the follow-up questions

Text-only baselines are character 2–5-gram TF-IDF features with logistic regression, scored out of fold (table 8). Look-alike made-up names join the first name of one obscure PopQA person to the surname of another (100 per person relation; the question template and options are as for the pseudo-word items). Date masking removes years, month names, weekdays, day numbers, seasons and phrases such as “by the end of” from the news questions (“Will the Fed cut rates by the end of March 2024?” → “Will the Fed cut rates?”); 6,274 of 6,320 questions change. With dates removed, post-change “no” answers fall from 92% to 67%; accuracy is reported only on answer-safe subsets (appendix C.2). When one sentence states the answer about a made-up subject, $P(\text{known})$ is 0.81, against 0.05 with an unrelated sentence of the same form and 0.20 with none, and Jev picks the stated option every time. Asked to verify both a proposed “yes” and a proposed “no” for every news item, Jev gives $P(\text{correct})$ 0.26 and 0.56 after the change point (0.40 and 0.54 before); with the negative wording (“is the proposed answer wrong?”) the implied answers agree with the positive wording on 95–96% of items, imply “yes” for 10% of post-change questions (true share 48%), and accept both proposals for 0.1%. The true share of “no” is 51.2% before and 51.5% after; the category mix differs, but accuracy falls within every news category, so shifts between broad categories alone do not explain the drop.

Table 8: AUROC for the known contrasts with and without surface cues. Each row uses one set of items (made-up / real, or after / before).

Contrast (items)	Text only	known	$1 - p_{\max}$	Second look
Pseudo-word made-up vs. all real (1,600 / 14,267)	1.000	0.911	0.746	—
Pseudo-word made-up vs. real, second-look subset (1,600 / 2,000)	—	0.916	0.747	0.80
Pseudo-word made-up vs. least popular real	1.000	0.833	0.685	—
Pseudo-word made-up vs. obscure real persons (600 / 555)	—	0.884	0.784	—
Look-alike made-up vs. obscure real persons (600 / 555)	0.492	0.741	0.761	—
News after vs. before, full panel (1,680 / 4,640)	0.997	0.913	0.389	—
News after vs. before, subset (1,680 / 1,680)	—	0.911	0.385	0.62
News, dates removed, same subset	0.831	0.580	0.558	—

Paired differences on the same items (known minus answer uncertainty $1 - p_{\max}$, bootstrap over items or months): realistic names −0.020 [−0.054, 0.013] (90% interval −0.048 to 0.007, inside ± 0.05); news with dates removed +0.023 [−0.012, 0.056] (90% interval −0.007 to 0.051); news with dates +0.526 [0.470, 0.578]. With text-classifier folds grouped by subject or month, the baselines are 1.000 (pseudo-word names), 0.498 (realistic names) and 0.831 (news with dates removed; 0.862 ungrouped).

C.2 Date removal and answer labels

Temporal expressions (years, month names, weekdays, day numbers, seasons and phrases such as “by the end of”) were removed together with their prepositions. A random sample of 240 (original, masked) pairs, 120 per period, was labelled against a written protocol by a single annotator, an AI assistant; a check by a second, human annotator is pending. Labels: *answer preserved* (the same event, and the original label holds under “has this happened by now?”; 83 pairs, all but one with label “yes”), *answer uncertain* (a deadline or window removed from a “no” question; 77) and *identity lost or ill-posed* (the date identified which occurrence, the question asked about a state at a time, or the edit broke it; 80). Dropping a deadline alone cannot turn a “yes” into a “no”, but 30 of the 112 “yes” pairs lost event identity, so “yes” questions are not all answer-safe. We therefore report only response changes: among all 1,648 “yes” questions, removing dates cut post-change “no” answers from 92% to 57% and pre-change ones from 41% to 37%. On the pairs judged answer-preserving: 92% to 65% after ($n = 37$), a drop of 0.27 [0.14, 0.43], with ten answers changing from “no” to “yes” and none the other way; 39% to 35% before ($n = 46$), a drop of 0.04 [−0.04, 0.13]. No accuracy on rewritten questions is used as label-valid. Setting today’s date to just after each event (the end of the month after the latest date the question mentions) leaves post-change accuracy at 0.514 and “no” answers at 91%.

Table 9: The “not known” option on dated news: coverage, accuracy on answered questions and confident errors (wrong with $p_{\max} \geq 0.9$).

Condition	Period	Coverage	Accuracy	Conf. errors (all)	Conf. errors (answered)
Base	before	100%	0.697	2.4%	2.4%
Base	after	100%	0.511	17.3%	17.3%
Today’s date given	after	100%	0.511	12.7%	12.7%
“Not known” allowed	before	64.3%	0.740	0.3%	0.5%
“Not known” allowed	after	6.6%	0.631	0.1%	0.9%
Date + “not known”	after	28.5%	0.491	0.1%	0.2%

C.3 Proper scoring rules

Table 10: Multiclass Brier score and log loss of the three-call mean distribution, with the values for a uniform answer over K options. Log loss uses $-\log \max(p, 10^{-4})$ without renormalisation, because Jev returns exact zeros; the floor therefore bounds, rather than measures, the loss on those items. With floors of 10^{-2} to 10^{-6} , MMLU-CF log loss ranges from 0.80 to 1.34 (5.8% of answers have probability zero on the correct option); TriviaQA (0.130–0.135) and dated news (0.709–0.711) barely change.

Set	K	n	Accuracy	Brier	Uniform	Log loss	Uniform
TriviaQA	4	9,960	0.955	0.067	0.750	0.132	1.386
PopQA (real subjects)	4	14,267	0.708	0.378	0.750	0.719	1.386
Daily Oracle yes/no	2	6,320	0.651	0.474	0.500	0.710	0.693
HotpotQA, non-supporting paragraphs	2	951	0.751	0.342	0.500	0.542	0.693
HotpotQA, both supporting paragraphs	2	951	0.958	0.071	0.500	0.193	0.693
MMLU-CF	4	10,000	0.776	0.377	0.750	1.075	1.386

For chance tasks, expected scores when outcomes follow the stated odds (ideal in parentheses): future draw with stated percentages, Brier 0.919 (0.608), log loss 2.36 (1.08); fair die, 1.321 (0.833) and 2.98 (1.79); loaded die, 1.000 (0.700) and 4.61 (1.50).

Table 11: Intent dose–response, all conditions (mean of three calls).

Code carries	Banking77				CLINC150			
	acc	$p_{\max}$	$\tilde{H}$	smECE	acc	$p_{\max}$	$\tilde{H}$	smECE
nothing	0.010	0.320	0.774	0.310	0.008	0.363	0.592	0.355
1 example	0.681	0.828	0.109	0.170	0.911	0.904	0.057	0.024
2 examples	0.796	0.879	0.074	0.099	0.958	0.953	0.028	0.019
4 examples	0.861	0.914	0.056	0.066	0.969	0.964	0.021	0.015
8 examples	0.899	0.927	0.046	0.037	0.979	0.974	0.016	0.016
name	0.821	0.893	0.069	0.084	0.926	0.933	0.037	0.025
name + 8	0.909	0.939	0.040	0.039	0.981	0.978	0.013	0.015
irrelevant text	0.019	0.312	0.559	0.293	0.007	0.226	0.579	0.219
swapped examples	0.001	0.932	0.044	0.904	0.001	0.974	0.016	0.957

Comparators. GLiNER2.5-Decide, given the same items and codes, is almost exactly uniform with no information ($p_{\max}$ 0.017 and 0.010), shows no first-code preference, and reduces entropy with knowledge, but is underconfident once informed (CLINC150 with names: $p_{\max}$ 0.34 at accuracy 0.66). A five-member MiniLM deep ensemble [23] trained on the same few examples shows mutual information falling with more data and stays cautious with one example ($p_{\max}$ 0.43–0.54) where Jev, more accurate, is confident (0.83–0.90).

Factual QA and abstention. Converted to four-option Choice: TriviaQA accuracy 0.955 (SmoothECE 0.031), SimpleQA Verified 0.753 (0.051), TruthfulQA binary 0.909 (0.041). Answering only at $p_{\max} \geq 0.9$ keeps 365 of 1,000 SimpleQA Verified questions with 1 error. As a verifier of a proposed TriviaQA answer, Jev’s yes/no probability reaches AUROC 0.984 and SmoothECE 0.028.

Unanswerable, ambiguous and disputed items. $P(\text{answerable})$ separates unanswerable SelfAware questions with AUROC 0.882; $P(\text{single answer})$ flags ambiguous AmbigQA questions only weakly (0.661). Jev’s entropy is rank-correlated with the entropy of 100 human labels on ChaosNLI ($\rho = 0.35$) but strongly compressed.

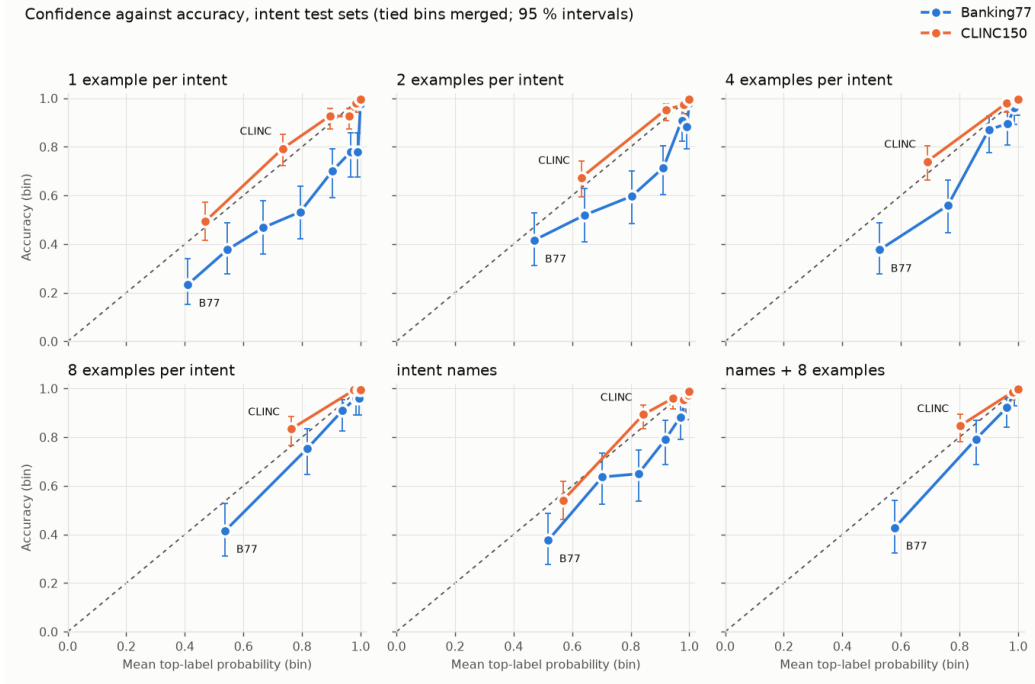


Figure 6: Intent dose–response reliability by what each intent code carries (about ten equal-count bins, with bins that would split identical confidence values merged; 95% intervals).

Table 12: Edge-of-knowledge settings. Gap = mean confidence – accuracy.

Setting	<i>n</i>	Accuracy	Confidence	Gap	SmoothECE
Fabricated entities (uniform reference 0.25)	1,600	—	0.518	—	—
Daily Oracle yes/no, after the change point	1,680	0.511	0.816	+0.305	0.305
MMLU-CF	10,000	0.776	0.914	+0.138	0.145
ANLI rounds 1–3	3,200	0.739	0.839	+0.100	0.105
HotpotQA, non-supporting paragraphs	951	0.751	0.830	+0.079	0.083
HotpotQA, one of two needed paragraphs	951	0.775	0.871	+0.096	0.104
Banking77, one example per intent	770	0.681	0.828	+0.147	0.170

Out-of-scope intents. An explicit out-of-scope option flags 89% of off-topic CLINC150 requests at 3.5% false alarms (AUROC 0.982), but only 36% of messages from held-out Banking77 intents with close relatives in the schema.

Random devices. Across all 720 option orders, Jev puts 0.80 on “one” for a fair die with number words, 0.74 on “1” with digits and 0.62 on “red” for colour faces; 1.00 on the 50% face of a loaded die; 0.97 on 7 for the sum of two dice. Mean probability by display position is flat, and averaging over orders leaves the distance to uniform unchanged (TV 0.632 vs. 0.634). With tied stated chances, the choice follows the order in which chances are mentioned (99.6%), not the display order.

Inferred settledness. For 360 made-up events in six domains, each in a past and a future version differing only in the cue, $P(\text{settled})$ separates past from future with AUROC 1.000 from tense alone (0.73 vs. 0.14), 0.996 from dates with today’s date given (0.68 vs. 0.30), and 0.898 from dates alone (0.64 vs. 0.53): Jev does not place itself in time unless told.

The “not known” option across mixes. With a share s of post-change questions (period-stratified month-block bootstrap, coverage matched exactly), the option minus a confidence threshold is -0.034 $[-0.045, -0.005]$ at $s = 0.05$, $+0.010$ $[-0.002, 0.043]$ at the date panel’s own share $s = 0.27$, and $+0.051$ $[0.016, 0.092]$ at $s = 0.5$; against a known threshold it is $+0.018$ to $+0.048$ at every share. Confident errors nearly vanish with the option (6 of 1,191 answers vs. 330), but this count is format-dependent (three-way vs. two-way distributions).

Wording robustness. Fourteen paraphrased, reversed and adversarial wordings of the three follow-up questions all rank the relevant contrast with AUROC 0.87–1.00, including adversarial settled cases that pit meaning against chance or future-tense words; absolute levels vary (for known, real subjects get 0.52–0.79 across wordings). For known these are the original, cue-bearing contrasts (appendix C.1).

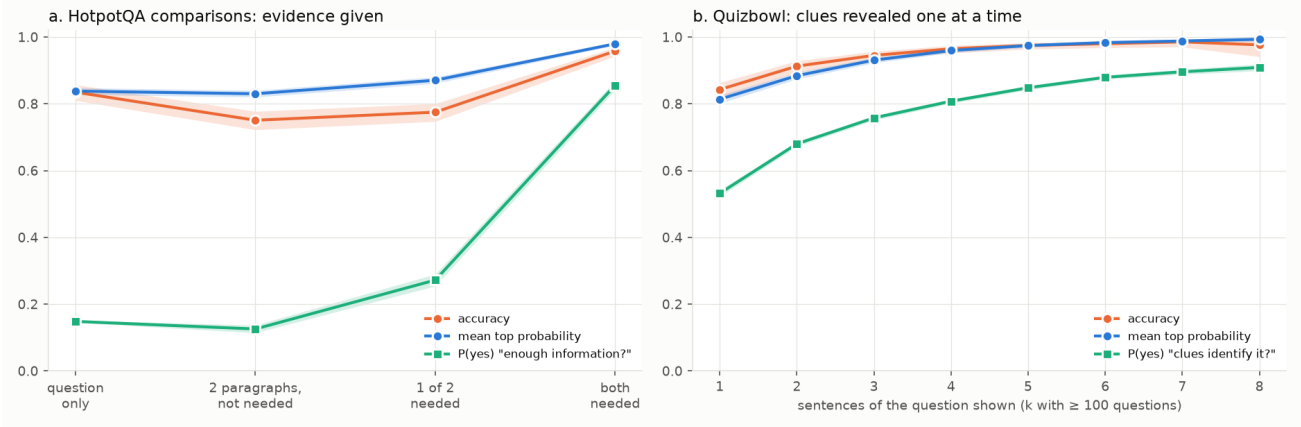


Figure 7: Evidence in context. (a) HotpotQA: supporting paragraphs raise accuracy and confidence together; non-supporting paragraphs reduce accuracy without a corresponding confidence drop, and partial evidence remains overconfident; $P(\text{enough})$ rises only with complete evidence. (b) Quizbowl: confidence and accuracy rise together clue by clue.

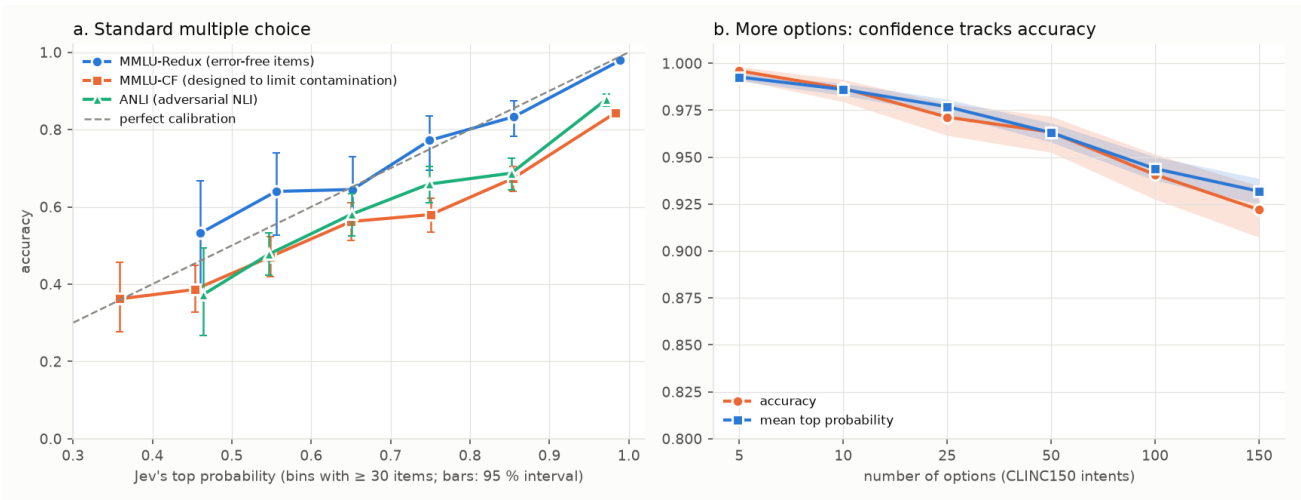


Figure 8: Standard multiple choice. (a) Reliability: MMLU-Redux on the diagonal; MMLU-CF and ANLI below it. (b) CLINC150 with 5–150 options.

D Experiment ledger

Generated from the response caches: every paid call is one cached record, attributed to the first experiment whose rebuilt requests include it. Units are item $\times$ condition combinations. The cache key includes the replicate index, so the replicates of a request are always separate paid calls; only requests identical in content *and* replicate index share a call, which happens when a later experiment reuses an earlier condition (counted under the earlier experiment) or when generated items coincide in text. The intent study's 27,240 units are the main arm (2,270 items $\times$ 9 conditions, reported in table 2) plus two robustness arms on 681 of those items: alternative example draws (7 conditions, two draws with one call each) and letter codes instead of numbered codes (3 conditions, 3 replicates); the development pilot used one call per unit, and two draws in the example arm. The instrument checks combine 129 API and instrument checks with 297 repeated-sampling calls, 24 of which are shared, giving 402. Paid calls are therefore not always units $\times$ replicates: the date / "not known" study's 13,440 units include 3,360 base-condition questions already paid under the knowledge boundary, so it adds $(13,440 - 3,360) \times 3 = 30,240$ calls. Of the 575,442 paid calls, 573,234 feed reported analyses; 2,208 are superseded or stray calls that are counted in the cost but not analysed. Dates are 2026 (UTC).

Table 13: Items, replicates and paid calls per experiment. *Planned before the run, plan committed afterwards (not externally verifiable).

Experiment	Units	Replicates	Paid calls	Dates	Planning
Instrument and sampling checks	—	varied	402	26 Sep	planned
Intent study, development pilot	6,356	1–2	7,945	26 Sep	pilot before the freeze
Intent dose–response (test)	27,240	2–3	76,953	26–28 Sep	plan frozen in a git tag
Out-of-scope detection	3,570	3 formats	10,710	28 Sep	planned
Knowledge $\times$ chance scenarios	6,000	3	18,000	28 Sep	planned*

Table 13: Items, replicates and paid calls per experiment. *Planned before the run, plan committed afterwards (not externally verifiable).

Experiment	Units	Replicates	Paid calls	Dates	Planning
Knowledge boundary (PopQA, fabricated, news)	26,927	3	80,781	28 Sep	planned*
Evidence in context (HotpotQA, Quizbowl)	10,622	3	31,866	28 Sep	planned*
Standard benchmarks	22,583	3	67,749	28 Sep	planned*
Wording robustness	3,080	3	9,231	28 Sep	planned*
Odds formats	1,440	3	4,320	28 Sep	planned*
Date / “not known” option	13,440	3	30,240	28 Sep	planned*
“Paragraphs may be irrelevant”	2,853	3	8,559	28 Sep	planned*
Standard multiple choice, option counts	27,900	3	83,604	28 Sep	planned*
Second look	15,523	3	46,566	28 Sep	planned*
Random devices	3,368	1–5	4,040	28 Sep	planned*
Policy cases	3,000	3	8,562	28 Sep	planned*
Open benchmark adaptations	6,663	3	19,989	28 Sep	planned*
Inferred settledness	2,880	3	8,640	29 Sep	written before the run
Dates removed	3,360	3	10,077	30 Sep	exploratory
Look-alike names	600	3	1,800	30 Sep	exploratory
Evidence for made-up subjects	800	3	2,400	30 Sep	exploratory
Opposite answer proposed	3,360	3	10,080	30 Sep	exploratory
Random devices, four wordings	480	1	480	30 Sep	exploratory
Negative verification wording	6,720	3	20,160	30 Sep	exploratory
Today’s date just after the event	3,360	3	10,080	30 Sep	exploratory
Superseded or stray calls	—	—	2,208	28–30 Sep	not used
Total			575,442		

E Datasets

Table 14: Public datasets, pinned revisions and licences.

Dataset	Source @ revision	Licence
Banking77 [9]	PolyAI task-specific-datasets @ 57ec275d	CC BY 4.0
CLINC150 [24]	clinc/clinc_oos @ 155b9c71	CC BY 3.0
PopQA [28]	akariasai/PopQA @ 098765c7	none stated
Daily Oracle [10]	agentic-learning-ai-lab/daily-oracle @ 455b35b2	CC BY 4.0
HotpotQA [48]	hotpotqa/hotpot_qa @ 1908d6af	CC BY-SA 4.0
Quizbowl [33]	community-datasets/qanta @ e3c56022	unknown
SelfAware [49]	ShuoZheLi/SelfAware @ fdfba5c	Apache-2.0
SimpleQA Verified [14]	google/simpleqa-verified @ 0dc97e0d	MIT
TruthfulQA [26]	truthfulqa/truthful_qa @ 741b8276	Apache-2.0
TriviaQA [18]	mandarjoshi/trivia_qa @ 0f7faf33	unknown
AmbigQA [29]	sewon/ambig_qa @ e969d013	CC BY-SA 3.0
ChaosNLI [32]	earino/chaosnli @ bf8d66a1	CC BY-NC 4.0
MMLU-Redux 2.0 [13]	edinburgh-dawg/mmlu-redux-2.0 @ 372ea425	CC BY 4.0
MMLU-CF [51]	microsoft/MMLU-CF @ c25b89a9	CDLA-Permissive-2.0
ANLI [31]	facebook/anli @ 8e4813d8	CC BY-NC 4.0

Free-text benchmarks were converted to four-option Choice: distractors are other objects of the same relation (PopQA), perturbed dates and numbers or the most similar answers of the same type (SimpleQA Verified), the most similar other answers (TriviaQA, Quizbowl), or the dataset’s own incorrect answer (TruthfulQA binary). Options are shown in seeded random order except MMLU, whose options keep their original order. The ChaosNLI and SelfAware mirrors were checked against independent copies. The six generated sets are: (1) 480 knowledge×chance scenarios; (2) fabricated-entity questions, 1,600 with pseudo-word names and 600 with look-alike names; (3) random devices (dice, coins, cards); (4) 600 rule-based policy cases; (5) 360 made-up events for inferred settledness; and (6) adversarial scenarios for the wording robustness of the follow-up questions.